# Morgan Huberty

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## *Education*

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**Johns Hopkins University** *Expected May 2027*

*Bachelor of Science in Computer Engineering*

Relevant Coursework: Mastering Electronics, Computer System Fundamentals, Signals & Systems

## *Projects*

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**Big Integers** *C++*

- Implemented a BigInt class (vector<uint64_t> magnitude, bool sign) supporting arbitrary-precision add, sub, mul, div, bit ops, and hex/dec conversion, with comprehensive unit tests

**Key/Value Store** *C++*

- Built a Redis-like in-memory K/V store with custom message serialization, multithreaded server (auto-commit & transactions with table-level locking), client utilities, and unit tests

**Cache Simulator** *C , C++*

- Simulated a configurable set-associative cache (size, block, LRU/FIFO, write policies) that outputs hit/miss and cycle counts from benchmark traces

**Parallel Merge Sort** *C*

- Implemented fork/join parallel merge sort on memory-mapped 64-bit integer files, with threshold-based sequential fallback, robust child-process management, and performance tuning experiments

## *Experience*

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**#iVoted Impact Concerts at SNF Agora Institute** *May 2025 - Present*

Data Science Researcher

- Analyzed music-listener and voter-demographic datasets (RedistrictingDataHub, Geocorr APIs) to inform #iVoted interventions
- Built projection dashboard for voter registrations using Python, Excel, and tracked data from #iVoted interventions

**National Society of Black Engineers** *September 2023 - Present*

Incoming Vice President; previous Finance Chair

- Managed financial planning for chapter activities, ensuring allocating and stewarding a $9K annual budget
- Developed marketing materials to promote member engagement and outreach by leveraging digital platforms
- Coordinated fundraising and sponsorship efforts for $3500 worth of income

**VEX Robotics** *November 2019 - March 2022*

Robotics Designer

- Designed and programmed competition-grade robotics platforms using C++ and Python
- Performed hardware bring-up (calibration; sensor integration) and basic circuit assembly (motor controllers; wiring harnesses)
- Collaborated with team members in maker-space environment, developing tools and strategies to optimize robot performance
- Troubleshot mechanical and electrical failures, documented design iterations and updated CAD models for 3D prints

## *Skills*

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**Programming Languages:** C, C++, Python (NumPy, Matplotlib, Pandas), Bash

*Familiarity with MATLAB, R, x86-64 Assembly (pixel graphics project)*

**Computer Applications:** Microsoft Suite, Google Suite, Google Sheets, Excel, Slack, Canva, Docker, Vercel, Git, Tinkercad, Fusion360

*Familiarity with KiCad, AWS, Google Cloud, Linux (Ubuntu), PyTorch, TensorFlow*